
Problem 8.22.

Part a. Let $ADD = \{\langle x, y, z\rangle \mid x, y, z > 0 \text{ are binary integers and } x + y = z\}$.
Show that $ADD \in$ L.

Proof. To show that $ADD \in$ L, we give a log space algorithm M that decides ADD.

$M =$ "On input $\langle x, y, z\rangle$, where x, y and z are binary integers greater than 0:

1. Let i, n_x, n_y, n_z and m be binary integers, and s and c be binary digits on work tape.
2. Let $n_x = |x|$, $n_y = |y|$, $n_z = |z|$, and m be maximum of n_x and n_y.
3. Let $i = 1$, and $c = 0$.
4. Repeat for m steps:
5. If z does not have i^{th} digit ($i > n_z$), then *reject*.
6. Let s be the sum of c and i^{th} binary digits of x and y. If either x or y does not have i^{th} binary digit, then assume the digit to be 0.
7. If the addition produces a carry, then $c = 1$. Otherwise, $c = 0$.
8. If s and i^{th} digit of z are different, then *reject*.
9. Increment i by 1.
10. If $c = 1$, then:
11. If z does not have i^{th} digit, then *reject*.
12. If c and i^{th} digit of z are different, then *reject*.
13. If z has $(i + 1)^{th}$ digit, then *reject*.
14. Otherwise, if z has i^{th} digit, then *reject*.
15. *Accept*."

□

Part b. Let $PAL - ADD = \{\langle x, y\rangle \mid x, y > 0 \text{ are binary integers where } x + y \text{ is an integer whose binary representation is a palindrome}\}$. (Note that the binary representation of the sum is assumed not to have leading zeros. A palindrome is a string that equals its reverse.)
Show that $PAL - ADD \in$ L.

Proof. □